THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–14) | Class 10 Mathematics | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 120	$2^3 \times 3 \times 5$
2	**(b)** 24	$2^3 \times 3$
3	**(a)** $4x^2 + x + 1$	$k(x^2 + \frac{1}{4}x + \frac{1}{4})$
4	**(b)** Incorrect — $x^2 + 4 > 0$, so it has no real zero	$x^2 + 4 \geq 4$
5	**(d)** 2	$\frac{3}{2k-1} = \frac{1}{k-1} \Rightarrow k = 2$; then $\frac{1}{5} \neq 1$
6	**(c)** parallel to the x-axis	All points have $y = -3$
7	**(b)** Incorrect — $D = 9 - 20 < 0$	Nature depends on D
8	**(c)** $x^2 - 5x + 6 = 0$	$D = 25 - 24 = 1 > 0$
9	**(c)** 5	$\frac{1}{2} + 9 \times \frac{1}{2}$
10	**(b)** $10, 20, 30, 40$	Add 10 each time
11	**(b)** 10 cm	$\frac{6}{BC} = \frac{3}{5}$
12	**(c)** RHS	RHS is a congruence rule
13	**(b)** $\frac{\sqrt{3}}{2}$	$2 \cdot \frac{1}{2} \cdot \frac{\sqrt{3}}{2}$
14	**(b)** 60°	Supplementary
15	**(b)** $\frac{22}{3}$ cm	$\frac{1}{6} \times 44$
16	**(b)** Student A is correct	Half of 616
17	**(d)** 462 cm²	$3\pi r^2$
18	**(c)** 6	$\frac{2+4+6+8+10}{5}$
19	**(b)** Incorrect — it is approximate (frequencies assumed at class marks)	Assumption
20	**(d)** $\frac{1}{6}$	1 of 6

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	If $\frac{2\sqrt{3}}{5} = r$ (rational), then $\sqrt{3} = \frac{5r}{2}$ would be rational — contradiction.	2
22	Sum 4, product $4 - 3 = 1 \Rightarrow x^2 - 4x + 1$	1+1
23	$(\sqrt{3}x - \sqrt{2})^2 = 0 \Rightarrow x = \sqrt{\frac{2}{3}}$ (twice)	1+1
24	$\frac{3}{6} = \frac{4}{8} = \frac{6}{12} = \frac{1}{2} \Rightarrow$ similar (SSS)	1+1
25	$(x-7)^2 + 36 = (x-3)^2 + 16 \Rightarrow 8x = 60 \Rightarrow x = \frac{15}{2} \Rightarrow \left(\frac{15}{2}, 0\right)$	1+1
26	$4\sin^2\theta + 3 = 4 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$, $\tan\theta = \frac{1}{\sqrt{3}}$	1+1
27	$\frac{6}{\sqrt{3}} = 2\sqrt{3} \approx 3.46$ m	2
28	$\sqrt{1681 - 81} = 40$ cm	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Points $(0,-8), (2,0)$; $(0,4), (2,0)$. Intersect at $(2,0) \Rightarrow$ consistent	2+1
30	$\Sigma fx = 20 + 60 + 150 + 120 + 100 = 450$, $\Sigma f = 30 \Rightarrow$ mean $= 15$	1+1+1
31	$n^2 - 37n + 342 = 0 \Rightarrow n = 18$ or 19; the 19th term is 0, so both sums are 513	1+1+1
32	$TA = TB \Rightarrow \angle TAB = 50^\circ$; $\angle OAB = 90^\circ - 50^\circ = 40^\circ$	1+1+1
33	35 min $= 210^\circ$: $\frac{210}{360} \times 1386 = 808.5$ cm²	1+1+1
34	$\frac{1}{36}$; $\frac{1}{2}$; $\frac{4}{36} = \frac{1}{9}$	1+1+1
35	$h\sqrt{3} - \frac{h}{\sqrt{3}} = 30 \Rightarrow h = 15\sqrt{3} \approx 25.98$ m	1+1+1

	OR: Building 20 m; total $20\sqrt{3}$ ⇒ flagpole $= 20(\sqrt{3}-1) \approx 14.64$ m	1+1+1
36	$\frac{980+60p}{20+p} = 50 \Rightarrow 10p = 20 \Rightarrow p = 2$	1+1+1
	OR: Modal class 60–80: $60 + \frac{16-12}{32-12-4} \times 20 = 65$	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $OA = 2$: $(x-2)^2 + 4 = 4 \Rightarrow x = 2$ (ii) $(0, 3)$ (iii) distance from O $= 2$ ⇒ yes	2+2+1
	OR: (i) $OF_1 = OF_2 = 50$ (ii) $\sqrt{78^2 + 26^2} = 26\sqrt{10} \approx 82.2$ m (iii) $(-9, 27)$	2+2+1
38	$= \frac{2\tan\theta\sec\theta}{\sec^2\theta - 1} = \frac{2\sec\theta}{\tan\theta} = \frac{2}{\sin\theta}$	2+2+1
	OR: Divide numerator and denominator by $\cos A$: $\frac{\frac{1}{\sin A} - 1}{\frac{1}{\sin A} + 1}$	2+2+1
39	(i) $432\pi - 288\pi = 144\pi \approx 452.57$ cm³ (ii) 3 : 2	3+2
	OR: (i) $231 + 89.83 = 320.83$ m³ (ii) $132 + 77 = 209$ m² (iii) ₹2090	2+2+1
40	$\Sigma f = 44 + f$, $\Sigma fx = 752 + 20f$ ⇒ $\frac{752+20f}{44+f} = 18 \Rightarrow f = 20$ (4); modal class 19–21 (1)	4+1
	OR: (i) Both $\frac{545}{25} = 21.8$ (ii) A: $20 + \frac{2.5}{10} \times 10 = 22.5$; B: $20 + \frac{2.5}{8} \times 10 = 23.125$ (iii) Section B	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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